

Random Forest



When One Tree Isn't Enough

Day 14 of 100 Days of Machine Learning

What is Random Forest?

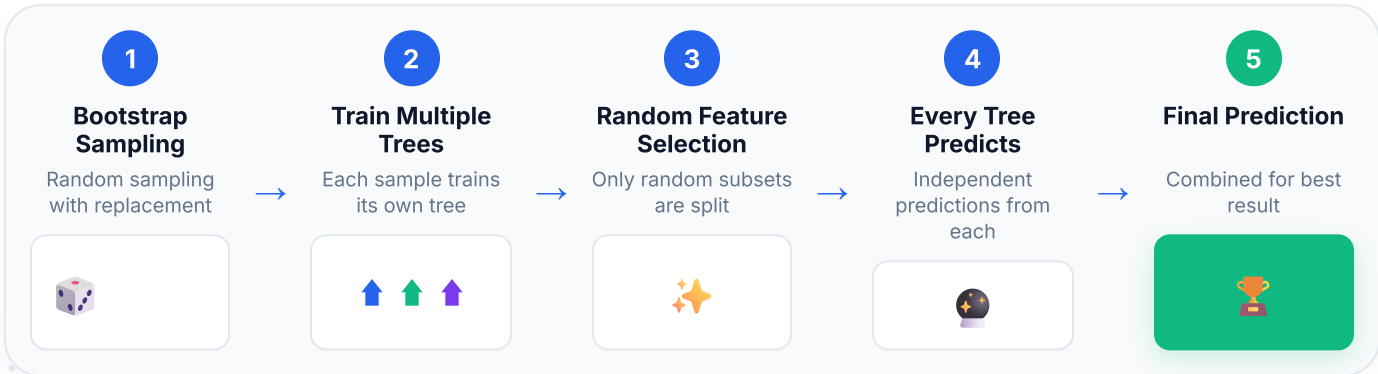
Random Forest is an ensemble learning algorithm that combines multiple decision trees to make better predictions. Instead of relying on one tree, it lets hundreds of trees **vote together**, making the model more accurate, stable, and resistant to overfitting.



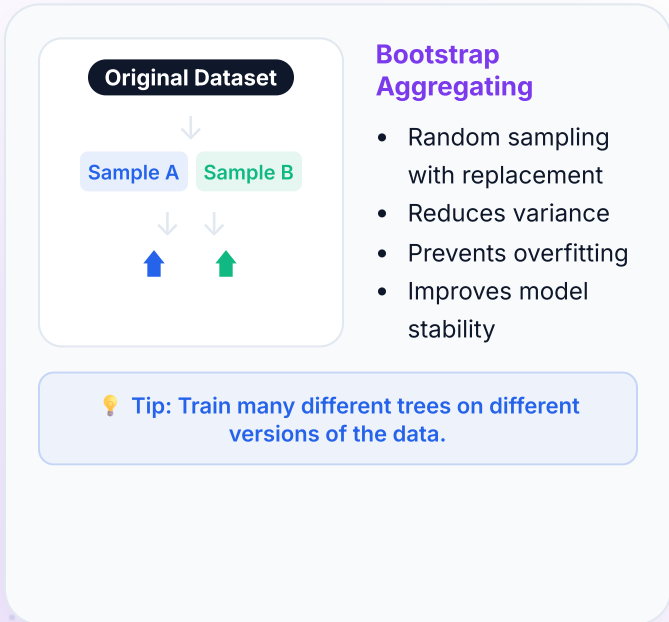
Why One Tree Isn't Enough



How Random Forest Works



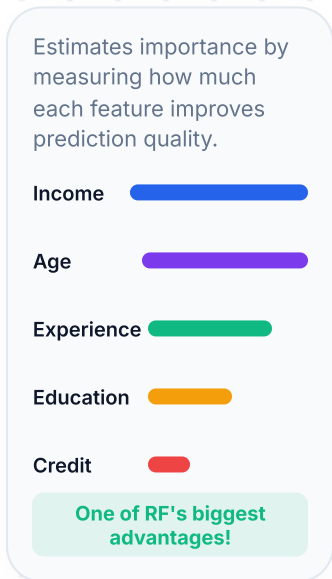
Bagging Explained



Voting Mechanism



Feature Importance



When To Use?

- ✓ High accuracy needed
- ✓ Nonlinear relationships

Advantages vs Disadvantages

- Advantages** ✓
 - ✓ High accuracy
 - ✓ Reduces overfitting
- Disadvantages** ✗
 - ✗ Slower than one tree
 - ✗ Harder to interpret

Hyperparameters

- n_estimators**
Number of trees
- max_depth**
Maximum depth